COURSE OUTLINE



DATA9401

Database Administration Fundamentals

Fall 2020 - Current

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DATA9401 Database Administration Fundamentals

COURSE DESCRIPTION

This course prepares you to take the Microsoft Technology Associate (MTA) exam: Database Fundamentals. This course introduces the fundamentals and core concepts of database development and administration, including the creation of database objects, manipulation of data, data storage, database security, backup, and recovery.

REQUISITES	None
EQUIVALENTS	None
CREDITS	0
HOURS	24
ELIGIBLE FOR PLAR	No

COURSE LEARNING OUTCOMES

Bow Valley College is committed to ensuring our graduates can demonstrate their abilities in key areas that will make them effective citizens and encourage their development as lifelong learners. In addition to the discipline-specific skills that learners acquire in their programs, the College has identified ten learning outcomes.

College-Wide Outcomes:

1. Communication
2. Thinking Skills
3. Numeracy and Financial Literacy
4. Working with Others
5. Digital Literacy
6. Positive Attitudes and Behaviours
7. Continuous Learning
8. Health and Wellness Awareness
9. Citizenship and Intercultural Competence
10. Environmental Sustainability

#	COURSE LEARNING OUTCOME(S)	COLLEGE WIDE OUTCOMES SUPPORTED
1	Explain core database and relational database concepts	Thinking Skills (2), Digital Literacy (5)
2	Create Tables and Views using SPSS and T-SQL	Thinking Skills (2), Digital Literacy (5)
3	Using Queries to Insert, Update, Modify and Delete from Database using SPSS and T-SQL	Thinking Skills (2), Digital Literacy (5)
4	Use Stored Procedures and Functions	Thinking Skills (2), Digital Literacy (5)
5	Perform Normalization	Thinking Skills (2), Digital Literacy (5)
6	Explain Database Security Concepts	Thinking Skills (2), Digital Literacy (5)

7	Understand and perform database backup and restore techniques	Thinking Skills (2), Digital Literacy (5)

COURSE MODULES AND SCHEDULE

**Course schedule subject to change, depending on delivery mode and term of study. For exact dates, please consult the Course Offering Information in Brightspace.*

WEEK/HOURS	MODULES
Week 1	Introduction • Read Syllabus and Outline • Familiarization of GMetrix and Microsoft SQL Server. Module 1 – Database Core Concepts • Read through Module 1 • Refer to the D2L learning Modules
Week 2	Module 1 – Database Core Concepts • Read through Module 1 • Refer to the D2L learning Modules. Quiz 1
Week 3	Module 2 – Datatypes, Table, View. Stored Procedures & Functions • Read through Module 2 • Refer to the D2L learning Modules
Week 4	Module 2 – Datatypes, Table, View. Stored Procedures & Functions • Read through Module 2 • Refer to the D2L learning Modules. Quiz 2
Week 5	Module 3 – SQL Queries • Read through Module 3 • Refer to the D2L learning Modules
Week 6	Module 3 – SQL Queries • Read through Module 3 • Refer to the D2L learning Modules
Week 7	Module 3 – SQL Queries • Read through Module 3 • Refer to the D2L learning Modules. Quiz 3
Week 8	Module 4 – Normalization and Indexes • Read through Module 4 • Refer to the D2L learning Modules
Week 9	Module 4 – Normalization and Indexes • Read through Module 4 • Refer to the D2L learning Modules
Week 10	Module 4 – Normalization and Indexes • Read through Module 4 • Refer to the D2L learning Modules. Quiz 4
Week 11	Module 5 – Database Security Concepts • Read through Module 5 • Refer to the D2L learning Modules
Week 12	Module 5 – Database Security Concepts • Read through Module 5 • Refer to the D2L learning Modules
Week 13	Module 5 – Database Security Concepts • Read through Module 5 • Refer to the D2L learning Modules. Quiz 5
Week 14	Exam Preparation & MTA Practice Exam
Week 15	Final MTA Exam

ASSESSMENT

COURSE LEARNING OUTCOME(S)	ASSESSMENT	WEIGHT
1	Quiz 1	10%
2	Quiz 2	10%
3	Quiz 3	10%
4, 5	Quiz 4	10%
6, 7	Quiz 5	10%
1, 2, 3, 4, 5, 7	Final MTA Exam	50%

Important: For details on each assignment and exam, please see the Course Offering Information.

PERFORMANCE STANDARDS

A grade of P is required to pass this course. Learners may be required to pass this course in order to progress in the program or to meet specific program completion requirements.

Please consult with the program area or the Bow Valley College website for further details.

GRADING SCHEME

Symbol	Description	Value in GPA
P	Pass	N/A
F	Fail	N/A

REQUIRED LEARNING RESOURCES

Additional learning resources may be found in the Course Offering Information or in Brightspace.

ADDITIONAL INFORMATION

Additional information may be found in the Course Offering Information or in Brightspace.

ACADEMIC ACCOMMODATIONS

Learners with a disability (learning, physical, and/or mental health) may qualify for academic and exam accommodations. For more information, or to apply for accommodations, learners should make an appointment with Accessibility Services in the Learner Success Services (LSS) Department. Accessibility Services can also assist learners who may be struggling with learning but do not have a formal diagnosis. To make an appointment visit LSS on the first floor of the south campus or call 403-410-1440. It is the learner's responsibility to contact Accessibility Services and request academic accommodations. For more information, please visit our website at http://www.bowvalleycollege.ca/accessibility.

INSTITUTIONAL POLICIES

Bow Valley College is committed to the highest standards of academic integrity and honesty. Learners are urged to become familiar with and uphold the following policies: Academic Honesty (500-1-7), Learner Code of Conduct, Procedures and Guidelines (500-1-1), Learner Appeals (500-1-12), Attendance (500-1-10), Grading (500-1-6), Academic Continuance and Graduation (500-1-5), and Electronic Communications (300-2-13). Audio or video recording of lectures, labs, seminars, or any other teaching and learning environment by learners is allowed only with consent of the instructor as

part of an approved accommodation plan. Recorded material is to be used solely for personal study and is not being used or distributed without prior written consent from the instructor.

Turnitin:

Students may be required to submit their course work to Turnitin, a third-party service provider engaged by BVC. Turnitin identifies plagiarism by checking databases of electronic books and articles, archived webpages, and previously submitted student papers. Students acknowledge that any course work or essays submitted to Turnitin will be included as source documents in the Turnitin.com reference database, where it will be used solely to detect plagiarism. The terms that apply to a student's use of Turnitin are described on Turnitin.com.

Online Exam Proctoring:

Examinations for this course may require proctoring through an online proctoring service. Online proctoring enables online exam taking within a controlled and monitored environment, thereby enhancing academic integrity. Online proctoring may occur through a variety of methods, including but not limited to:

a. live online proctoring where a remote invigilator authenticates identity and observes completion of an exam using specialized software and recordings;
b. automated proctoring where the exam session is recorded and AI (artificial intelligence) analyzed;
c. browser lockdown that limits access to other applications, websites, copying, printing, screen capture and other functions; or
d. a combination of both live/automated proctoring and browser lockdown.

Course instructors will review recordings, analyses, and data obtained through online proctoring for academic integrity infractions. It is the student's responsibility to meet the technical, software, location, and identity verification requirements necessary to enable online proctoring.

Further details of these policies are available in the Academic Calendar and on the Bow Valley College website, bowvalleycollege.ca.

Learners are encouraged to keep a copy of this course outline for future reference.